Algebra
Home Work 6

Student: 26-712-224

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"I wrote these solutions fully myself und would like to receive feedback on these"

A) Let p be a prime number. We will give an easy proof of the fact that every group of order p^2 is abelian using that every nontrivial p-group has a nontrivial center. (Compare with Exercise B of Sheet 4.) Let G be a p-group of order p^2 and assume that G is non-abelian. Obtain a contradiction as follows.

(1) Show that $Z(G)$ is cyclic of order p.
(2) Fix $g \in G - Z(G)$. Show that $\mathrm{ord}(g) = p$.
(3) Show that $\langle g \rangle \cap Z(G) = \{1\}$.
(4) Show that $\langle g \rangle \cdot Z(G) = G$.
(5) Deduce that G is abelian, and conclude.

B) Let G be a finite group acting on a finite set X. For $g \in G$, we set

$$X^g = \{x \in X \mid g \cdot x = x\}.$$

This is the subset of elements of X fixed by g. The goal of the exercise is to prove the following formula, due to Burnside:

$$\mathrm{card}(G\backslash X) = \sum_{g \in G} \frac{\mathrm{card}(S^g)}{\mathrm{card}(G)}. \qquad (\star)$$

Step 1. We first consider the case where $X = G/H$, for a subgroup $H < G$, endowed with the natural action given by

$$\begin{aligned} G \times G/H &\longrightarrow G/H \\ (g, xH) &\longmapsto gxH. \end{aligned}$$

To treat this case, we introduce the set

$$E = \{(g, k) \in G \times G \mid k^{-1}gk \in H\}.$$

We have two maps $p_1, p_2 : E \to G$ given by $p_1(g, k) = g$ and $p_2(g, k) = k$.

(1) Show that p_2 is surjective. For $k \in G$, show that there is a bijection from $p_2^{-1}(k)$ to the subgroup $k \cdot H \cdot k^{-1}$. Deduce that

$$\mathrm{card}(E) = \mathrm{card}(H) \cdot \mathrm{card}(G).$$

(*Hint:* Use that E is the disjoint union of the $p_2^{-1}(k)$'s.)

(2) Let $g \in G$. Show that there is a bijection from $p_1^{-1}(g)$ to the set

$$\begin{aligned} F_g &= \{k \in G \mid g \cdot (kH) = kH\} \\ &= \{k \in G \mid kH \in (G/H)^g\}. \end{aligned}$$

Use that the H-cosets belonging to $(G/H)^g$ form a partition of the set $\{k \in G \mid kH \in (G/H)^g\}$ to obtain the equality

$$\mathrm{card}(F_g) = \mathrm{card}((G/H)^g) \cdot \mathrm{card}(H).$$

Deduce that

$$\mathrm{card}(E) = \sum_{g \in G} \mathrm{card}((G/H)^g) \cdot \mathrm{card}(H).$$

(3) Use (1) and (2) to obtain the Burnside formula $(\star)$ for $X = G/H$ endowed with the natural action of G.

Step 2. We now prove the general case of the Burnside formula. Let $O_1, \cdots, O_r$ be the orbits of the action of G on X. (Thus $G\backslash X = \{O_1, \cdots, O_r\}$.)

(4) Note that the group G acts on every O_i. Show that S^g is the disjoint union of the $(O_i)^g$'s. Deduce that it suffices to prove the Burnside formula for each one of the orbits O_i endowed with its natural action of G.

(5) Let $s \in O_i$ and set $H = \mathrm{Stab}_G(s)$. Using the bijection $G/H \xrightarrow{\sim} O_i$, deduce a bijection between $(O_i)^g$ and $(G/H)^g$. Use Step 1 to conclude.